

Early Warning System & Prediction API

API Documentation • Version 1.0.0

Student ERP — Risk, Dropout, GPA, Fee, Admission predictions and Early-Warning alerts

Overview

This API exposes machine-learning style predictions and an early-warning alert system for a student management ERP. All prediction endpoints use the **POST** method and exchange **JSON**. Send the header Content-Type: application/json with every POST request.

Base URL (local): `http://127.0.0.1:8000`

Interactive docs: `/docs` (Swagger UI) • `/redoc` • `/openapi.json`

Risk levels

Every 0–1 score maps to a human-friendly risk level:

Score range	risk_level
0.00 – 0.29	low
0.30 – 0.59	medium
0.60 – 0.79	high
0.80 – 1.00	critical

Endpoints

POST**/prediction/student-risk**

Predicts whether a student is at academic risk and may fail. Returns a risk score, risk level, a may_fail flag and any triggered alerts.

Parameters

Field	Type	Required	Description
student_id	string	true	Unique student identifier
attendance_percentage	number (0–100)	true	Attendance percentage
current_gpa	number (0–4)	true	Current GPA
assignments_submitted	integer	true	Assignments submitted
assignments_total	integer	true	Total assignments
backlogs	integer	no	Failed/pending subjects (default 0)

Example request (JSON body)

```
{
  "student_id": "STU-1001",
  "attendance_percentage": 62,
  "current_gpa": 2.1,
  "assignments_submitted": 6,
  "assignments_total": 10,
  "backlogs": 2
}
```

Example response (200 OK)

```
{
  "student_id": "STU-1001",
  "risk_score": 0.419,
  "risk_level": "medium",
  "may_fail": false,
  "alerts": [ { "type": "low_attendance", "severity": "medium", ... } ]
}
```

POST**/prediction/dropout**

Predicts the probability that a student drops out, with risk level and alerts.

Parameters

Field	Type	Required	Description
student_id	string	true	Unique student identifier
attendance_percentage	number (0–100)	true	Attendance percentage
current_gpa	number (0–4)	true	Current GPA
fee_overdue_days	integer	no	Days fee is overdue (default 0)
engagement_score	number (0–1)	true	Class participation 0–1

Example request (JSON body)

```
{
  "student_id": "STU-1001",
  "attendance_percentage": 55,
  "current_gpa": 1.8,
  "fee_overdue_days": 30,
  "engagement_score": 0.3
}
```

Example response (200 OK)

```
{
  "student_id": "STU-1001",
  "dropout_probability": 0.61,
  "risk_level": "high",
  "alerts": [ ... ]
}
```

POST**/prediction/gpa**

Predicts a student's future GPA (0–4) together with a confidence value.

Parameters

Field	Type	Required	Description
student_id	string	true	Unique student identifier
previous_gpa	number (0–4)	true	Previous GPA
attendance_percentage	number (0–100)	true	Attendance percentage
study_hours_per_week	number	true	Weekly study hours
assignments_submitted	integer	true	Assignments submitted
assignments_total	integer	true	Total assignments

Example request (JSON body)

```
{
  "student_id": "STU-1001",
  "previous_gpa": 3.0,
  "attendance_percentage": 85,
  "study_hours_per_week": 12,
  "assignments_submitted": 9,
  "assignments_total": 10
}
```

Example response (200 OK)

```
{
  "student_id": "STU-1001",
  "predicted_gpa": 3.25,
  "confidence": 0.96
}
```

POST**/prediction/fee-default**

Predicts the probability that a student defaults on fee payment.

Parameters

Field	Type	Required	Description
student_id	string	true	Unique student identifier
outstanding_amount	number	true	Outstanding fee amount
days_overdue	integer	no	Days overdue (default 0)
previous_late_payments	integer	no	Past late payments (default 0)
total_payments	integer (≥ 1)	true	Total payments made so far

Example request (JSON body)

```
{
  "student_id": "STU-1001",
  "outstanding_amount": 45000,
  "days_overdue": 20,
  "previous_late_payments": 3,
  "total_payments": 8
}
```

Example response (200 OK)

```
{
  "student_id": "STU-1001",
  "default_probability": 0.41,
  "risk_level": "medium",
  "alerts": []
}
```

POST**/prediction/admissions**

Predicts whether an applicant should be admitted. Decision is admit (≥ 0.70), waitlist (≥ 0.50) or reject (< 0.50).

Parameters

Field	Type	Required	Description
applicant_id	string	true	Unique applicant identifier
entry_test_score	number (0–100)	true	Entry test score
previous_grade_percentage	number (0–100)	true	Previous academic %
interview_score	number (0–100)	no	Interview score (default 0)

Example request (JSON body)

```
{
  "applicant_id": "APP-2025-077",
  "entry_test_score": 78,
  "previous_grade_percentage": 82,
  "interview_score": 70
}
```

Example response (200 OK)

```
{
  "applicant_id": "APP-2025-077",
  "admission_probability": 0.77,
  "decision": "admit"
}
```

POST**/prediction/recommendations**

Generates personalised recommendations for a student based on attendance, GPA and weak subjects.

Parameters

Field	Type	Required	Description
student_id	string	true	Unique student identifier
attendance_percentage	number (0–100)	true	Attendance percentage
current_gpa	number (0–4)	true	Current GPA
weak_subjects	array of string	no	Subjects the student is weak in

Example request (JSON body)

```
{
  "student_id": "STU-1001",
  "attendance_percentage": 60,
  "current_gpa": 2.2,
  "weak_subjects": ["Math", "Physics"]
}
```

Example response (200 OK)

```
{
  "student_id": "STU-1001",
  "recommendations": [
    "Improve attendance to at least 75% ...",
    "Attend extra tutoring for Math.", ...
  ]
}
```

GET**/alerts/student/{student_id}**

Early Warning System + ERP integration. Fetches a student from the ERP, runs all models and returns every triggered alert. In mock mode try STU-1001 (risky) or STU-1002 (healthy).

Parameters

Field	Type	Required	Description
student_id	string (path)	true	Student id in the URL path
push_to_erp	boolean (query)	no	Also save alerts back into the ERP (default false)

Example response (200 OK)

```
[
  {
    "student_id": "STU-1001",
    "type": "low_attendance",
    "severity": "medium",
    "message": "Attendance is 62%, below the required 75%",
    "score": 0.473
  }
]
```

Validation: missing fields or out-of-range values return **422 Unprocessable Entity** with details. Unknown student on the alerts endpoint returns **404 Not Found**. Validation is automatic via FastAPI + Pydantic.